

YOUNG JIN YOO'S CURRICULUM VITAE

Postdoctoral researcher
Research Laboratory of Electronics (RLE)
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RESEARCH INTERESTS

Membrane-based integrated photonics; multi-physics simulation-driven design-to-fab

Silicon does not host the materials that modulate, emit, or store light. Those materials are released as single-crystalline freestanding membranes, transferred onto CMOS-compatible platforms, and the resulting devices are designed and fabricated in-house or by CMOS foundry tape-out.

- ♦ Complex-oxide membranes (LiNbO₃, BaTiO₃): PLD growth, crystallographic orientation and interface engineering, DFT-guided defect and oxygen-vacancy control
- ♦ Electro-optic modulators built on photonic-crystal cavities, nanobeams, and microrings
- ♦ III-V and III-N on-chip lasers and resonant-cavity LEDs, integrated as coupons at high density
- ♦ Heterogeneous integration onto CMOS and photonic ICs by remote and 2D-material-assisted epitaxy and membrane transfer
- ♦ Applications: optical interconnects and co-packaged optics for computing; resonant-cavity LED microdisplays; quantum photonic integrated circuits
- ♦ Design-to-fab loop: scope the target and the benchmark to beat, design the device and its fabrication window by coupled, cross-validated multi-physics simulation inside the CMOS design rules (with inverse design and AI/ML surrogates), build it with DFT-guided interface engineering and membrane transfer, and prove it by device and optical measurement read out with coupled-mode theory. Each pass calibrates the model and de-risks the next fabrication run.

WORK EXPERIENCE

Postdoctoral researcher	Massachusetts Institute of Technology (MIT), US Research Laboratory of Electronics, Mar. 2023 ~ Present <i>Co-advised by Prof. Jeehwan Kim and Prof. Dirk R. Englund</i> <i>- Membrane-based active photonics: electro-optic modulators on complex-oxide membranes; III-V and III-N lasers and resonant-cavity LEDs integrated onto CMOS-compatible platforms</i> <i>- Photonic integrated circuit testing and heterogeneous device integration for quantum information science (AFRL/RITQ)</i> <i>- Multi-physics design-to-fab: device and fabrication-window definition by coupled simulation, CMOS foundry tape-out, and measurement read-out</i>
Visiting scholar	Georgia Institute of Technology (GT), US George W. Woodruff School of Mechanical Engineering, Jul. 2022 ~ Oct. 2022
Postdoctoral researcher	Gwangju Institute of Science and Technology (GIST), Korea School of Electrical Engineering and Computer Science, Mar. 2021 ~ Feb. 2023

EDUCATION

Ph.D. Degree	Gwangju Institute of Science and Technology (GIST), Korea School of Electrical Engineering and Computer Science, Mar. 2018 ~ Feb. 2021 Thesis: Ultra-Thin Films Colorimetric Sensors with Highly Absorbing Media (advisor: Prof. Young Min Song)
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M.S. Degree	Gwangju Institute of Science and Technology (GIST), Korea School of Electrical Engineering and Computer Science, Sep. 2016 ~ Feb. 2018 Thesis: Study on Ultra-Thin Film Coloration with Highly Absorbing Media (advisor: Prof. Young Min Song)
B.S. Degree	Pusan National University, Korea Department of Electronics Engineering, Mar. 2010 ~ Feb. 2016

AWARDS AND FELLOWSHIPS

Sejong Science Fellowship by National Research Foundation of Korea (Mar. 2021 – Feb. 2024)

Encouragement prize in the 26th Humantech Paper Award by Samsung electronics, Korea (11 Feb. 2020)

Title: “Ultra-thin porous films with highly absorbent media for colorimetric detection”, **Y. J. Yoo**, J. H. Ko, Y. J. Kim, and Y. M. Song

Global Ph.D Fellowship by National Research Foundation of Korea (Mar. 2018 – Feb. 2021)

Excellent Paper Award in the Photonics Conference, Korea (9 Nov. 2017)

Title: “Optimal design of anti-reflective nanostructures with enhanced hardness by using atomic layer deposition”, **Y. J. Yoo**, G. J. Lee, J. H. Lee, J. W. Lee, C. S. Lee, and Y. M. Song

Best Poster Award in the International Symposium on Green Manufacturing and Applications, Korea (29 Jun. 2017)

Title: “Ultra-thin film optical coatings for reflective color filters”, **Y. J. Yoo**, and Y. M. Song

Scholarship supported by Korean government (Sep. 2016 – Feb. 2021)

Excellent Paper Award in the Photonics Conference, Korea (27 Nov. 2014)

Title: “Design of ZnS antireflective microstructures for mid- and far-infrared applications”, **Y. J. Yoo**, K. S. Chang, and Y. M. Song

ARCHIVAL JOURNALS

43. B. Kim, S. H. Choi, G. Zograf, **Y. J. Yoo**, Y. Baek, S. Kim, J. Kim, J. Kim, S. H. Chae, H. Kim, S. Hong, and K. Lee, *Co-packaged optics for high-performance computing and artificial intelligence*, Nat. Electron. 9, 853–867 (2026). **Co-first author**

42. J. H. Ko, H. E. Jeong, S. Kim, D. Kim, S. Y. Kim, **Y. J. Yoo**, H.-H. Jeong, and Y. M. Song, *Sub-1-volt, reconfigurable Gires-Tournois resonators for full-coloured monapixel array*, Light Sci. Appl. 15, 134 (2026).

41. S. Lee, X. Zhang, P. Abdollahi, M. Barone, C. Dong, **Y. J. Yoo**, M.-K. Song, D. Lee, J.-E. Ryu, J.-H. Choi, J.-H. Lee, J. A. Robinson, D. G. Schlom, H. S. Kum, C. S. Chang, A. Seo, and J. Kim, *Route to Enhancing Remote Epitaxy of Perovskite Complex Oxide Thin Films*, ACS Nano, 18, 31225–31233 (2024).

40. J. H. Ko, J. -E. Yeo, H. E. Jeong, H. M. Kim, **Y. J. Yoo**, Y. Yuk, S. Lee, and Y. M. Song, *Switchable and Conspicuous Retroreflective Sensors Inspired by the Wing Scale of an Emerald Swallowtail*, Biosens. Bioelectron. 260, 116445 (2024).

39. P. Prabhathan, K. V. Sreekanth, J. Teng, J. H. Ko, **Y. J. Yoo**, H.-H. Jeong, Y. Lee, S. Zhang, T. Cao, C.-C. Popescu, B. Mills, T. Gu, Z. Fang, R. Chen, H. Tong, Y. Wang, Q. He, Y. Lu, Z. Liu, H. Yu, A. Mandal, Y. Cui, A. S. Ansari, V. Bhingardive, M. Kang, C. K. Lai, M. Merklein, M. J. Müller, Y. M. Song, Z. Tian, J. Hu, M. Losurdo, A. Majumdar, X. Miao, X. Chen, B. Gholipour, K. A. Richardson, B. J. Eggleton, M. Wuttig, and R. Singh, *Roadmap for phase change materials in photonics and beyond*, iScience 26, 107946 (2023).

38. J. H. Ko, J.-H. Park, **Y. J. Yoo**, S. Chang, J. Kang, A. Wu, F. Yang, S. Kim, H.-G. Jeon, Y. M. Song, *Full-Control and Switching of Optical Fano Resonance by Continuum State Engineering*, Adv. Sci. 10, 2304310 (2023). **Co-first author**
37. J. Kang, **Y. J. Yoo**, J.-H. Park, J. H. Ko, S. Kim, S. G. Stanciu, H. A. Stenmark, J. Lee, A. A. Mahmud, H.-G. Jeon, Y. M. Song, *DeepGT: Deep learning-based quantification of nanosized bioparticles in bright-field micrographs of Gires-Tournois biosensor*, Nano Today 52, 101968 (2023). **Co-first author**
36. H. Kim, **Y. J. Yoo**, J. H. Yun, S. -Y. Heo, Y. M. Song, W. -H. Yeo, *Outdoor Worker Stress Monitoring Electronics with Nanofabric Radiative Cooler-Based Thermal Management*, Adv. Healthcare Mater. 12, 2301104 (2023). **Co-first author**
35. J. H. Yun, **Y. J. Yoo**, H. R. Kim, and Y. M. Song, *Recent progress in thermal management for flexible/wearable devices*, Soft Sci 3, 12 (2023). **Co-first author**
34. J. Kang, **Y. J. Yoo**, J. H. Ko, A. A. Mahmud, and Y. M. Song, *Trilayered Gires–Tournois Resonator with Ultrasensitive Slow-Light Condition for Colorimetric Detection of Bioparticles*, Nanomaterials 13, 319 (2023). **Co-first author**
33. J. H. Ko, S. H. Kim, M. S. Kim, S. -Y. Heo, **Y. J. Yoo**, Y. J. Kim, H. Lee, and Y. M. Song, *Lithography-Free, Large-Area Spatially Segmented Disordered Structure for Light Harvesting in Photovoltaic Modules*, ACS Appl. Mater. Interfaces 14, 44419–44428 (2022).
32. J. H. Ko, **Y. J. Yoo**, Y. Lee, H.-H. Jeong, and Y. M. Song, *A review of tunable photonics: Optically active materials and applications from visible to terahertz*, iScience, 25, 104727 (2022). **Co-first author**
31. J.-K. Song, J. Kim, J. Yoon, J. H. Koo, H. Jung, K. Kang, S.-H. Sunwoo, S. Yoo, H. Chang, J. Jo, W. Baek, S. Lee, M. Lee, H. J. Kim, M. Shin, **Y. J. Yoo**, Y. M. Song, T. Hyeon, D. -H. Kim, D. Son, *Stretchable colour-sensitive quantum dot nanocomposites for shape-tunable multiplexed phototransistor arrays*, Nat. Nanotech., 17, 849–856 (2022).
30. H. M. Kim, **Y. J. Yoo**, J. M. Lee, and Y. M. Song, *A Wide Field of View Light Field Camera with Adjustable Multiplicity for Practical Applications*, Sensors, 22, 3455 (2022). **Co-first author**
29. W. Lee, **Y. J. Yoo**, J. Park, J. H. Ko, Y. J. Kim, H. Yun, D. H. Kim, Y. M. Song, and D. -H. Kim, *Perovskite microcells fabricated using swelling-induced crack propagation for colored solar windows*, Nat. Comm. 13, 1946 (2022). **Co-first author**
28. **Y. J. Yoo**, J. H. Ko, G. J. Lee, J. Kang, M. S. Kim, S. G. Stanciu, H. -H. Jeong, D. -H. Kim, and Y. M. Song, *Gires-Tournois immunoassay platform for label-free bright-field imaging and facile quantification of bioparticles*, Adv. Mater. 34, 2110003 (2022). **(Front Cover Picture Article)**
27. **Y. J. Yoo**, Y. M. Song, *Editorial for the Topic on Micromachining for Advanced Biological Imaging*, Micromachines. 13, 474 (2022).
26. S. H. Kim, J. H. Ko, **Y. J. Yoo**, M. S. Kim, G. J. Lee, S. Ishii, and Y. M. Song, *Single-Material, Near-Infrared Selective Absorber Based on Refractive Index-Tunable Tamm Plasmon Structure*, Adv. Opt. Mater. 10, 2102388 (2022). **(Inside Front Cover Picture Article)**
25. J. H. Lee, Y. J. Kim, **Y. J. Yoo**, S. Chang, G. J. Lee, J. H. Ko, K. M. Kang, D. Chanda, and Y. M. Song, *Colored, Covert Infrared Display through Hybrid Planar-Plasmonic Cavities*, Adv. Opt. Mater. 9, 2100429 (2021). **(Inside Front Cover Picture Article)**
24. **Y. J. Yoo**, S.-Y. Heo, Y. J. Kim, J. H. Ko, Z. F. Mira and Y. M. Song, *Functional photonic structures for external interaction with flexible/wearable devices*, Nano Res., 14, 2904–2918 (2021).

23. K. J. Lee, J.-W. Min, B. Turedi, A. Y. Alsalloum, J.-H. Min, Y. J. Kim, **Y. J. Yoo**, S. Oh, N. Cho, R. C. Subedi, S. Mitra, S. E. Yoon, J. H. Kim, K. Park, T.-H. Chung, S. H. Jung, J.-H. Baek, Y. M. Song, I. S. Roqan, T. K. Ng, B. S. Ooi, and O. M. Bakr, *Nanoporous GaN/n-type GaN: a cathode structure for ITO-free perovskite solar cells*, ACS Energy Lett., 5, 3295–3303 (2020).
22. **Y. J. Yoo**, W.-G. Kim, J. H. Ko, Y. J. Kim, Y. Lee, S. G. Stanciu, J.-M. Lee, S. Kim, J.-W. Oh, and Y. M. Song, *Large-Area Virus Coated Ultrathin Colorimetric Sensors with a Highly Lossy Resonant Promoter for Enhanced Chromaticity*, Adv. Sci. 7, 2000978 (2020). ([Inside Back Cover Picture Article](#))
21. **Y. J. Yoo**, J. H. Ko, W.-G. Kim, Y. J. Kim, D.-J. Kong, S. Kim, J.-W. Oh, and Y. M. Song, *Dual-Mode Colorimetric Sensor Based on Ultrathin Resonating Facilitator Capable of Nanometer-Thick Virus Detection for Environment Monitoring*, ACS Appl. Nano. Mater. 3, 6636–6644 (2020).
20. Y. J. Kim, **Y. J. Yoo**, M. H. Kang, J. H. Ko, M. R. Park, D. E. Yoo, D. W. Lee, K. Kim, I. -S. Kang, and Y. M. Song, *Mechanotunable optical filters based on stretchable silicon nanowire arrays*, Nanophotonics 9, 3287–3293 (2020). **Co-first author**
19. S. G. Stanciu, D. E. Tranca, L. Pastorino, S. Boi, Y. M. Song, **Y. J. Yoo**, S. Ishii, R. Hristu, F. Yang, G. Bussetti, and G. A. Stanciu, *Characterization of Nanomaterials by Locally Determining their Complex Permittivity with Scattering-Type Scanning Near Field Optical Microscopy*, ACS Appl. Nano Mater. 3, 1250–1262 (2020).
18. J. H. Ko, **Y. J. Yoo**, Y. J. Kim, S.-S. Lee and Y. M. Song, *Flexible, Large-Area Covert Polarization Display Based on Ultrathin Lossy Nanocolumns on a Metal Film*, Adv. Funct. Mater. 30, 1908592 (2020) ([Front Cover Picture Article](#)). **Co-first author**
17. **Y. J. Yoo**, Y. J. Kim, S. -Y. Kim, J. H. Lee, K. Kim, J. H. Ko, J. W. Lee, B. H. Lee, and Y. M. Song, *Mechanically robust antireflective moth-eye structures with a tailored coating of dielectric materials*, Opt. Mater. Express 9, 4178 (2019).
16. Y. J. Kim, **Y. J. Yoo**, D. E. Yoo, D. W. Lee, M. Kim, H. J. Jang, Y. -C. Kim, J. -H. Jang, I. -S. Kang, and Y. M. Song, *Enhanced light harvesting in photovoltaic devices using an edge-located one-dimensional grating polydimethylsiloxane membrane*, ACS Appl. Mater. Interfaces 11, 36020–36026 (2019).
15. D. H. Kim, **Y. J. Yoo**, J. H. Ko, Y. J. Kim, and Y. M. Song, *Standard red green blue (sRGB) color representation with a tailored dual-resonance mode in metal/dielectric stacks*, Opt. Mater. Express 9, 3342 (2019). **Co-first author**
14. H. J. Jang, Y. J. Kim, **Y. J. Yoo**, G. J. Lee, M.S. Kim, K. S. Chang, and Y. M. Song, *Double-Sided Anti-Reflection Nanostructures on Optical Convex Lenses for Imaging Applications*, Coatings 9, 404 (2019).
13. Y. J. Kim, **Y. J. Yoo**, G. J. Lee, D. E. Yoo, D. W. Lee, V. Siva, H. S. Song, I. S. Kang, and Y. M. Song, *Enlarged Color Gamut Representation Enabled by Transferable Silicon Nanowire Arrays on Metal–Insulator–Metal Films*, ACS Appl. Mater. Interfaces 11, 11849–11856 (2019).
12. H. S. Song, G. J. Lee, D. E. Yoo, Y. J. Kim, **Y. J. Yoo**, D. W. Lee, V. Siva, I. S. Kang, and Y. M. Song, *Reflective color filter with precise control of the color coordinate achieved by stacking silicon nanowire arrays onto ultrathin optical coatings*, Sci. Rep. 9, 3350 (2019)
11. H. M. Kim, M. S. Kim, G. J. Lee, **Y. J. Yoo** and Y. M. Song, *Large area fabrication of engineered microlens array with low sag height for light-field imaging*, Opt. Exp. 27, 4435 (2019).
10. G. J. Lee, Y. J. Kim, H. M. Kim, **Y. J. Yoo**, Y. M. Song, *Colored, Daytime Radiative Coolers with Thin-Film Resonators for Aesthetic Purposes*, Adv. Opt. Mater. 6, 1800707 (2018) ([Front Cover Picture Article](#)).

9. Y. J. Kim, G. J. Lee, S. K. Kim, J. W. Min, S. Y. Jeong, **Y. J. Yoo**, S. H. Lee, Y. M. Song, *Efficient Light Absorption by GaN Truncated Nanocones for High Performance Water Splitting Applications*, ACS Appl. Mater. Interfaces 10, 28672–28678 (2018).
8. G. J. Lee, **Y. J. Yoo**, and Y. M. Song, *Recent advances in imaging systems and photonic nanostructures inspired by insect eye geometry*, Appl. Spectrosc. Rev. 53, 112–128 (2018).
7. H. S. Song, **Y. J. Yoo**, G. J. Lee, K. S. Chang, and Y. M. Song, *Optical Design of Porous ZnO/TiO₂ Films for Highly Transparent Glasses with Broadband Ultraviolet Protection*, J. Nanomater. 2017, 2738015 (2017). **Co-first author**
6. **Y. J. Yoo**, G. J. Lee, K. I. Jang, and Y. M. Song, *Fabrication of Ultra-thin Color Films with Highly Absorbing Media Using Oblique Angle Deposition*, J. Vis. Exp. 126, e56383 (2017).
5. **Y. J. Yoo**, J. H. Lim, G. J. Lee, K.-I. Jang, and Y. M. Song, *Ultra-thin films with highly absorbent porous media fine-tunable for coloration and enhanced color purity*, Nanoscale 9, 2986–2991 (2017) ([Front Cover Picture Article](#)).
4. J.-K. Song, D. H. Son, J. M. Kim, **Y. J. Yoo**, G. J. Lee, L. Wang, M. K. Choi, J. W. Yang, M. C. Lee, K. S. Do, J. H. Koo, N. Lu, J. H. Kim, T. H. Hyeon, Y. M. Song, and D.-H. Kim, *Wearable Force Touch Sensor Array Using a Flexible and Transparent Electrode*, Adv. Func. Mater. 27, 1605286 (2017).
3. W. I. Nam, **Y. J. Yoo**, and Y. M. Song, *Geometrical shape design of nanophotonic surfaces for thin film solar cells*, Opt. Exp. 24, A1033 (2016). **Co-first author**
2. **Y. J. Yoo**, K. S. Chang, and Y. M. Song, *Design of ZnO hollow nanosphere arrays for UV absorbing transparent glasses*, Opt. Quant. Electron. 48, 88 (2016).
1. **Y. J. Yoo**, K. S. Chang, S. W. Hong and Y. M. Song, *Design of ZnS antireflective microstructures for mid- and far-infrared applications*, Opt. Quant. Electron. 47, 1503–1508 (2015).

RESEARCH PROJECTS

TOPCHIP: Photonic Integrated Circuit Testing and Heterogeneous Device Integration for Quantum Information Science

Feb. 2026 - present

Funding: Quantum Information Sciences Branch, Air Force Research Laboratory Information Directorate (AFRL/RITQ), US Air Force (BAA FA8750-23-S-7001)

Principal investigator: Prof. Dirk R. Englund

Individual role: photonic integrated circuit testing and heterogeneous device integration; design and fabrication of membrane-based active photonic devices, carried out jointly between the groups of Prof. Dirk R. Englund and Prof. Jeehwan Kim.

Pre-patterned Freestanding Single-crystalline Lithium Niobate Photonic Components for Advanced Quantum Photonic Integrated Circuits

Sep. 2024 – present

Funding: Electronics, Photonics and Magnetic Devices (EPMD) of the National Science Foundation (NSF), US

Principal investigator: Prof. Jeehwan Kim

Individual role: preparation of research proposal, staff in charge of project, development of pre-patterned lithium niobate photonic components.

Development of an Entire Life Implantable Nanomesh Brain-Machine Integration Interface for the Neurological Disorders

Jul. 2024 – present

Funding: STEAM Research Business – Korea Global Cooperative Convergence Research Program of the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT,

Korea

Principal investigator: Prof. Jeehwan Kim

Individual role: preparation of research proposal, staff in charge of project, development of neuronal-resolution micro-LEDs array.

AI deep learning-based virus colorimetric detection on Gires-Tournois immunoassay platform

Mar. 2022 – Feb. 2023

Funding: AI-based social problem-solving convergence talent nurturing support project of Gwangju Institute of Science and Technology (GIST), Korea

Principal investigator: Prof. Young Min Song

Individual role: preparation of research proposal, staff in charge of project, development of Gires-Tournois immunoassay platform.

Development of active meta material and device with complex light modulation for high definition and wide viewing angle hologram

Jul. 2021 – Feb. 2023

Funding: the nano material technology development project funded by the Ministry of Science and ICT, Korea

Principal investigator: Prof. Young Min Song

Individual role: preparation of research proposal, design of complex optical modulation active meta-atoms for high-definition wide viewing angle hologram.

Airborne virus colorimetric sensing smart skin based on ultra-thin film platform

Mar. 2021 – Feb. 2024

Funding: Sejong Science Fellowship of the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT, Korea

Principal investigator: Dr. **Young Jin Yoo**

Individual role: preparation of research proposal, staff in charge of project, development of smart skin system.

Light field and LiDAR sensor fusion systems for full self-driving

Apr. 2020 – Dec. 2021

Funding: ICT R&D innovation voucher support project funded by the Ministry of Science and ICT, Korea

Principal investigator: Prof. Yong Min Song

Individual role: preparation of research proposal, design of wide field of view light field cameras.

Ultra-thin films colorimetric sensors with highly absorbent media

Mar. 2018 – Feb. 2021

Funding: Global PhD Fellowship of the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT, Korea

Principal investigator: **Young Jin Yoo**

Individual role: preparation of research proposal, staff in charge of project, development of ultra-thin films colorimetric sensors.

Development of visual enhancement material based on bacteriophage metamaterials

Sep. 2017 – present

Funding: creative materials discovery program of the National Research Foundation of Korea (NRF) funded by the Ministry of Science and ICT, Korea

Principal Investigator: Prof. Young Min Song

Individual Role: preparation of research proposal, staff in charge of project, development of dynamic color films for visual enhancement.

Integrated approaches of physically unclonable cryptographic primitives using random lasers and optoelectronics

Jun. 2017 – Jun. 2020

Funding: Korea-US air force science Lab joint research program by Institute for Information & communications Technology Promotion (IITP) grant funded by the Korea government (MSIP)

Principal investigator: Prof. Young Min Song

Individual role: preparation of research proposal, development of Zinc oxide random lasers.

Nano-textured transparent plastic for anti-reflection

May. 2017 – Aug. 2018

Funding: Hyundai NGV of Hyundai Motor Group, Korea

Principal investigator: Prof. Young Min Song

Individual role: preparation of research proposal, staff in charge of project, development of nano-textured transparent film for anti-reflection.

PROCEEDINGS AND PRESENTATIONS (INTERNATIONAL)

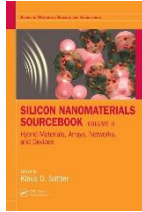
15. **Y. J. Yoo**, W.-G. Kim, J. H. Ko, Y. Lee, J.-W. Oh, and Y. M. Song, *Virus-based ultra-thin film colorimetric sensors for enhanced chromaticity*, Conference on Lasers and Electro-Optics (CLEO) 2020, ATu4I.2 (Virtual conference, May 2020)
14. **Y. J. Yoo**, J. H. Ko, Y. J. Kim, and Y. M. Song, *Covert polarization display based on ultra-thin lossy nanocolumns with wide color selectivity*, Conference on Lasers and Electro-Optics (CLEO) 2020, AF3I.2 (Virtual conference, May 2020)
13. J. H. Ko, **Y. J. Yoo**, and Y. M. Song, *Polarization Sensitive Ultra-thin Color Filter with Highly Structured Nano-column*, Frontiers in Optics (FiO) 2019, FM3F.2 (Washington, D.C., United States, Sep. 2019).
12. D. H. Kim, **Y. J. Yoo**, J. H. Ko, Y. J. Kim, and Y. M. Song, *Enhanced color purities for additive colors enabled by 1D metal-insulator resonator*, Frontiers in Optics (FiO) 2019, JW3A.115 (Washington, D.C., United States, Sep. 2019).
11. **Y. J. Yoo**, and Y. M. Song, *Ultrathin films-based colorimetric detector with highly absorbent porous media*, International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE) 2018, 2-2-2410 (Jeju, Korea, Nov. 2018).
10. Y. J. Kim, H. S. Song, G. J. Lee, **Y. J. Yoo**, Vantri Siva, and Y. M. Song, *Fine tuning color coordinate with stacking polymer embedded vertical sparse nanowire arrays for reflective color filter applications*, International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE) 2018, 2-2-2470 (Jeju, Korea, Nov. 2018).
9. **Y. J. Yoo**, and Y. M. Song, *Hydrophilic ultra-thin porous films for color variation with humidity by using oblique angle deposition*, Material Science Engineering (MSE) 2018, P11.4-4 (Darmstadt, Germany, Sep. 2018).
8. **Y. J. Yoo**, and Y. M. Song, *Ultra-thin porous films for colorimetric detection with highly absorbing media*, International Symposium on Sensor Science (ISS) 2018, 020161 (Kenting, Taiwan, Aug. 2018).
7. Y. J. Kim, G. J. Lee, **Y. J. Yoo**, and Y. M. Song, *GaN anti-reflective nanostructure for enhancing solar water splitting efficiency*, Materials Research Society (MRS) 2018, EN11 (Phoenix, United States, Apr. 2018)
6. **Y. J. Yoo**, and Y. M. Song, *Fine Tuning of Reflective Colors by Introducing Porosity in Ultra-thin Lossy Films*, Frontiers in Optics (FiO) 2017, JT2A.117 (Washington, D.C., United States, Sep. 2017).
5. **Y. J. Yoo**, and Y. M. Song, *Ultra-thin film optical coatings for reflective color filters*, International Symposium on Green Manufacturing and Applications (ISMGA) 2017, PP179 (Gyeongju, Korea, Jun. 2017).
4. **Y. J. Yoo**, W. I. Nam, and Y. M. Song, *Antireflective property on nanophotonic surface of different*

- shapes in thin-film solar cells*, The International conference & Exhibition for Nanotechnology (NANOPIA) 2016, DP-P19 (Changwon, Korea, Nov. 2016).
3. **Y. J. Yoo**, and Y. M. Song, *Design of ZnO multilayers with different porosities for UV absorbing transparent glasses*, International Conference on Numerical Simulation of Optoelectronic Devices (NUSOD) 2016, MP22 (Sydney, Australia, Jul. 2016).
 2. M. S. Kim, **Y. J. Yoo**, G. J. Lee and Y. M. Song, *Nanoscale 3D patterning with a phase mask*, International Symposium on Nature-Inspired Technology (ISNIT) 2016, S1-P10 (Daejeon, Korea, Jan. 2016).
 1. **Y. J. Yoo**, J. H. Lee, and Y. M. Song, *Design of hollow silica nanospheres for high efficiency optical devices applications*, International Conference on Numerical Simulation of Optoelectronic Devices (NUSOD) 2015, MP31 (Taipei, Taiwan, Sep. 2015).

PATENTS

10. J.-E. Ryu, J. Kim, **Y. J. Yoo**, and J. H. Ko, Method of Enhancing Light Extraction Efficiency in Resonant Cavity LEDs via Interface and Polarization Engineering, US provisional patent application No. 63/942,903 (Dec. 2025).
9. J. H. Ko, **Y. J. Yoo**, H.-H. Jeong, and Y. M. Song, *Hologram implementation device using metamaterial*, Korea patent pending No. 10-2021-0172256 (Dec. 2021).
8. **Y. J. Yoo**, J. H. Ko, J. H. Lee, and Y. M. Song, *Color based virus detector*, Korea patent pending No. 10-2021-0053637 (Apr. 2021).
7. M. S. Kim, H. M. Kim, **Y. J. Yoo**, J. H. Lee, H. J. Jang, and Y. M. Song, *Vehicle with light field camera*, Korea patent pending No. 10-2020-0163724 (Nov. 2020).
6. **Y. J. Yoo**, J. H. Ko, W.-G. Kim, J.-W. Oh, and Y. M. Song, *Large-area ultra-thin film color developing structure and method of preparing thereof*, Korea patent pending No. 10-2019-0168773 (Dec. 2019).
5. **Y. J. Yoo**, J. H. Ko, and Y. M. Song, *Method for ultra-thin large area hidden image generation*, Korea patent pending No. 10-2019-0112918 (Sep. 2019).
4. G. J. Lee, Y. J. Kim, **Y. J. Yoo**, and Y. M. Song, *Passive radiating cooling structure*, Korea patent pending No. 10-2019-0057225 (May. 2019).
3. Y. J. Kim, G. J. Lee, **Y. J. Yoo**, and Y. M. Song, *Coloring structure deposited polymer thin film and manufacturing method for polymer thin film*, Korea patent pending No. 10-2019-0024393 (Feb. 2019).
2. **Y. J. Yoo**, W.-G. Kim, J. M. Lee, J.-W. Oh, and Y. M. Song, *Porous ultra-thin film color developing structure and method of manufacturing the same*, Korea patent No. 10-2185557 (Nov. 2020).
1. **Y. J. Yoo**, G. Ju, and Y. M. Song, *Coloring structure and method for manufacturing coloring structure*, Korea patent No. 10-1961688 (Mar. 2019), PCT pending No. PCT/KR2018/002513 (Mar. 2018).

BOOK CHAPTERS



Klaus D. Sattler et al., 'Silicon Nanomaterials Sourcebook', CRC press

Chapter 15. Antireflective silicon nanostructures, **Y. J. Yoo**, E. K. Kang, Y. T. Lee and Y. M. Song

ACADEMIC SERVICES

Editorial board member for **Frontiers in Photonics**

Reviewer for Science Advances, Nano Letters, Nano Energy, IEEE Journal of Selected Topics in Quantum Electronics, IEEE Photonics Journal, Journal of Luminescence, Soft Science, Micromachines, Electronics

TECHNICAL EXPERTISE

Semiconductor processing of GaAs, GaN, Si, Ge, and other related materials

- Holographic lithography, photo-lithography, plasma-enhanced chemical vapor deposition, RF magnetron sputtering with ellipsometer, reactive ion etching, inductively coupled plasma etching, wet thermal oxidation, electron-beam evaporation, rapid thermal annealing, wafer lapping and polishing, wafer dicing and wire bonding

Thin-film growth and interface engineering

- Pulsed laser deposition (PLD) of complex oxides: BaTiO₃ and LiNbO₃ on CMOS-compatible templates
- Interface engineering to set the crystallographic orientation required for the electro-optic response
- DFT-guided defect and oxygen-vacancy engineering of oxide films and their interfaces

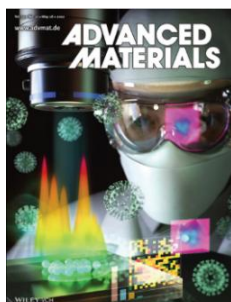
Characterization tools

- Scanning electron microscopy, atomic force microscopy, surface profiler
- Photoluminescence, UV-VIS-NIR spectrum measurement, Hall measurement, *in-situ* reflectance monitoring of thin film
- L-I-V measurement for LDs and LEDs, optical spectrum analysis, RF spectrum analysis, small signal modulation, eye diagram measurement
- Solar simulator, Incident photon-to-current efficiency (IPCE) measurement
- Water contact angle measurement, humidity, and chemical gas sensing measurement

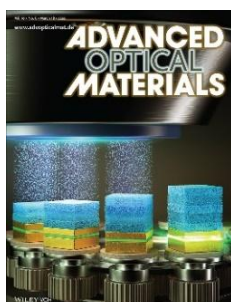
Programming / Simulation

- Photonics: FDTD, RCWA, mode solving, ray optics; photonic-crystal band structure and cavity design
- Devices and materials: TCAD (devices), DFT (materials), FEM (electrostatics, mechanics)
- Inverse design and AI/ML surrogate modeling; coupled-mode theory for measurement read-out
- Python-scripted workflows with cloud-based solvers; Matlab
- RSoft (DiffractionMod, FullWave), Zemax OpticStudio, Essential Macleod

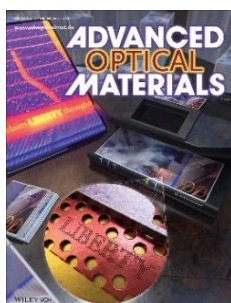
COVER PICTURES



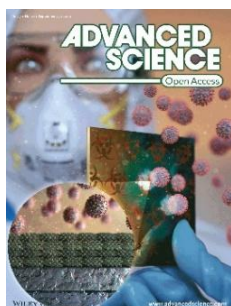
The cover image shows a concept illustration of Gires–Tournois immunoassay platform for label-free bright-field imaging and facile quantification of bioparticles. Gires–Tournois immunoassay platform (GTIP) for label-free bright-field imaging and facile quantification of bioparticles is proposed. Bioparticles on GTIP slow down the velocity of reflected light, leading to vivid color change according to the local particle density and maximizing chromatic contrast. The particle distribution and density are analyzed through 2D raster-scanning–based chromaticity analysis. **Adv. Mater.** **34**(21), 2022



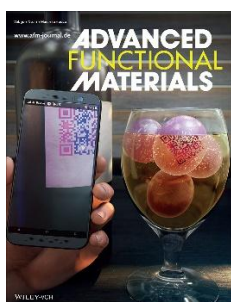
The cover image shows a concept illustration of single-material, near-infrared selective absorber based on refractive index-tunable Tamm plasmon structure. A design flow for single-material sub-micron planar Tamm plasmons (TPs) with near-unity absorptance is presented, where the refractive indexes are tailored through porosity tuning to achieve impedance matching. TPs with high absorptance ($\approx 99\%$) and Q-factors (≈ 45) in the Tamm state are realized. As a practical demonstration, a single material TP metal semiconductor metal photodetector is fabricated and an ultra-sensitive narrowband photodetection is realized. **Adv. Opt. Mater.** **10**(6), 2022



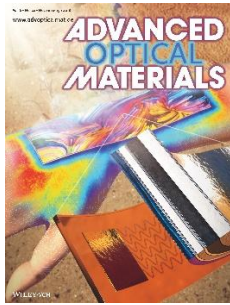
The cover image shows a concept illustration of colored, covert infrared display through hybrid planar-plasmonic cavities. A hybrid planar-plasmonic cavity (HPPC) display, capable of the encoding capacity exceeding those available in conventional thermal anti-counterfeit systems, is presented. Optimized two spectrally distinguished resonant structures of the HPPC allow for developing esthetic/camouflage anti-counterfeit labels. HPPC theoretically confirms that colorations can be obtained using diverse material combinations. The flexible HPPC is successfully demonstrated in different applications for practical use. **Adv. Opt. Mater.** **9**(17), 2021



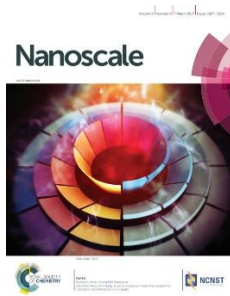
The cover image shows a concept illustration of virus coated ultrathin colorimetric sensors with a highly lossy resonant promoter. A large-area virus coated ultrathin colorimetric sensor with a highly lossy resonant promoter (HLRP) is developed. A dynamically responsive virus-based changeable coloring strategy with HLRP is an efficacious approach to fuse the advantages of strong resonant photonic structures and dynamically responsive materials. An ultrathin virus layer for rapid response to external stimuli displays colorimetric behavior on HLRP. **Adv. Sci.** **7**(18), 2020



The cover image shows a concept illustration of covert polarization display based on ultrathin lossy nanocolumns on a metal film. A flexible, large-area covert polarization display based on ultrathin lossy nanocolumns with wide color selectivity is developed. Self-aligned porous nanocolumns fabricated by glancing angle deposition represent a facile approach for the polarization of distinguishable structures. Covert polarization display labels are attached to daily objects with curved and wrinkled surfaces, and hidden quick response codes are revealed by polarization adjustment. **Adv. Funct. Mater.** **30**(11), 2020



The cover image shows a concept illustration of colored passive radiative cooler (CPRC). A colored passive radiative cooler (CPRC) capable of generating potential cooling power, based on a thin-film optical resonator embedded in an efficient thermal emission structure. This CPRC not only selectively emits infrared wave through the atmospheric transparency window but also displays subtractive primary colors to exhibit the desired appearance. Successful demonstration of cooling/coloring behavior with wearable electronic devices under solar irradiation represents a major step forward in the field of temperature-sensitive, flexible, wearable electronic/optoelectronic devices. **Adv. Opt. Mater. 6(22), 2018**



The cover image shows a concept illustration of ultra-thin color films with highly absorbent media. Ultra-thin, fine-tunable optical coatings with enhanced color purity are demonstrated based on highly absorbent porous media on a metal substrate. The color range provided by these ultra-thin film coatings can be extended by making the absorptive dielectric layer porous. Oblique angle deposition (OAD) of a thin germanium (Ge) film by e-beam evaporation onto a thick gold substrate yields controlled porosity. Angle independent reflection properties, and the applicability of such porous Ge on various metal substrates, indicate the strength of these concepts. **Nanoscale 9, 2017**